realistic images where there are large areas of light and shade.

Contrast measurement – the specification game

Marketing people love numbers they can stretch or misrepresent to make their products look better and for displays, contrast ratio is one of those numbers. It would seem that measuring contrast should be straight forward, as mentioned contrast is the ratio between peak white and black. However, as the 'black' level falls to lower and lower amounts of light it becomes more and more difficult to measure accurately. It is possible to claim huge contrast ratios simply by measuring black levels in a very dark room with the display or at least the back-light, literally turned off.

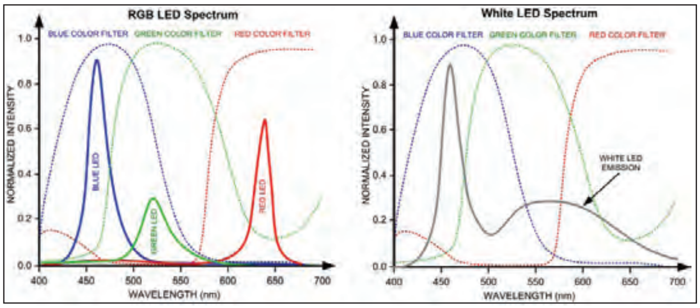
Measuring with a checker board pattern of black and white rather than with the entire screen black and white, allows for internal screen reflection and produces a more realistic result. As does measuring with a known amount of ambient light.

Typical contrast ratios for LCD panels are on the order of 200 to 400 to 1, although more recent developments in PVA (Patterned Vertical Alignment) and S-PVA (Super Patterned Vertical Alignment) panels provide claimed contrast ratios of over 3,000 to 1 (i.e. better black levels). The old CRTs provided good contrast because, when properly adjusted, black level means the display emits no light.

Colour stability and temperature compensation

All colour displays have suffered from colour instability due to temperature change. With CRT and the CCFL back lit LCD panels the approach, certainly before colour calibrating a display, was to leave the display switched on and displaying an image for between half an hour, to an hour, to allow it to 'warm up' and for the colour to stabilise. There was no attempt in these designs to apply corrective feedback for colour/temperature changes.

The light output of LEDs does vary markedly with ambient temperature and in an RGB LED back-light this can result in colour drift as the temperature characteristics of red, green and blue LEDs do differ.



Spectral plot of an RGB back lit display compared to a white LED back lit display. The RGB plot shows clear peaks for red, green and blue while the white LED plot has a sharp peak at around 450nM and a lower, broader, peak from 500 to 600nM – i.e. blue and yellow

White LED back-lights and comparative gamuts

Perhaps an obvious and relatively simple step in display design evolution is to take a conventional LCD panel and simply replace the Cold Cathode back-light with a white-light, LED back-light system. The advantage of doing this is that the electronics are simplified because the high voltage converter required by the CCFL isn't needed and the environmental impact is at least modified because LEDs don't contain mercury.

White LED back-light displays often do not provide the best in wide gamut photo-realism. This is because the spectral output of the the "white" light from the LEDs is not very smooth and does not provide sufficient energy over a wide range of frequencies. Most white LEDs are blue, or ultra-violet light LEDs, with the addition of a yellow phosphor in the lens cap, which converts some of the blue light into light at other frequencies, in the same way that a fluorescent tube phosphor converts the ultra-violet wavelengths into visible light. This is evident in the spectral plot (see – white LED spectral plot) where the blue shows as a sharp spike with a much lower broader yellow peak. Consequently the light from this type of 'white' LED is characteristically a hard blue-white light, due to the high blue peak. Just to confuse things, Samsung refer to the LED backlight technology they use as LED-BLU where BLU stands for Back Light Unit.

Typically, a blue plus yellow phosphor LED back-light will only manage a 71.6 per cent comparative gamut (relative to NTSC). Experiments have also been done using a blue LED with distinct red and green phosphors. When this LED design is combined with the typical LCD filter sets and colour matching functions, a reasonably wide gamut of 91.9 per cent of the NTSC standard gamut is obtained.

In comparison manufacturers using back-light designs with discrete red, green and blue LEDs are claiming gamuts as large as 123 per cent compared to the NTSC standard.

Unfortunately, companies offering LED back-light displays use different designs and don't want to reveal exactly how their products work, it may be difficult for the consumer to tell with certainty what the differences are between the various products available in order to make sensible choices. There is for example the significant difference between a 'white' LED back-light and an RGB back-light. A display may even use an RGB LED back-light and as a result offer a better colour gamut, but not apply segmented brightness control to provide the improvement in contrast. d